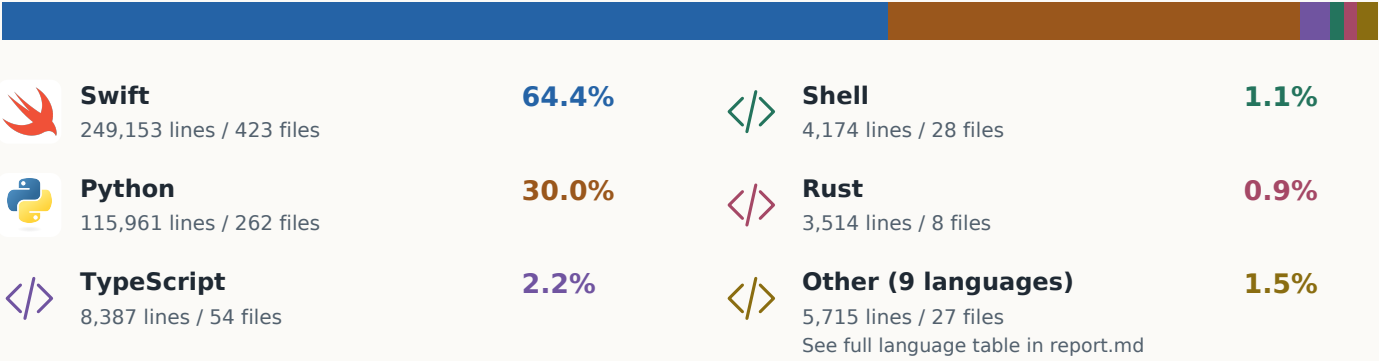


What is the code made of?

SOURCE COMPOSITION / Code volume above, technology responsibilities below

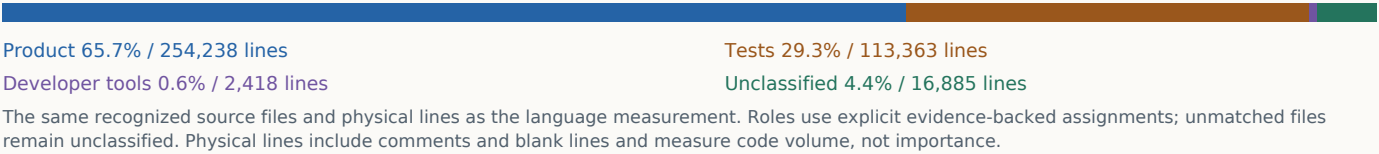
386,904 physical lines

802 source files / 14 languages



Physical lines include comments and blank lines. This measures code volume, not importance.

WHAT THE SOURCE IS FOR



WHAT EACH TECHNOLOGY DOES

**INTERFACE**
SwiftUI

Builds native workspace panels.
Used / confirmed / E6, E55

**SERVER AND API**
FastAPI

Composes domain API routes and agent transport.
Used / confirmed / E3, E93

**AI PROVIDER**
Ollama

Streams the selected local model in the traced Ollama branch.
Used / confirmed / E69, E70

**LANGUAGE RUNTIME**
Python

Runs the local agent backend.
Configured / confirmed / E91, E92

**NATIVE INTEGRATION**
EventKit

Reads and creates permitted calendar events.
Used / confirmed / E22, E23

**DATABASE**
SQLite

Stores run metadata, checkpoints and schedule records. Chat transcripts are separate JSONL files.
Used / confirmed / E94-E96

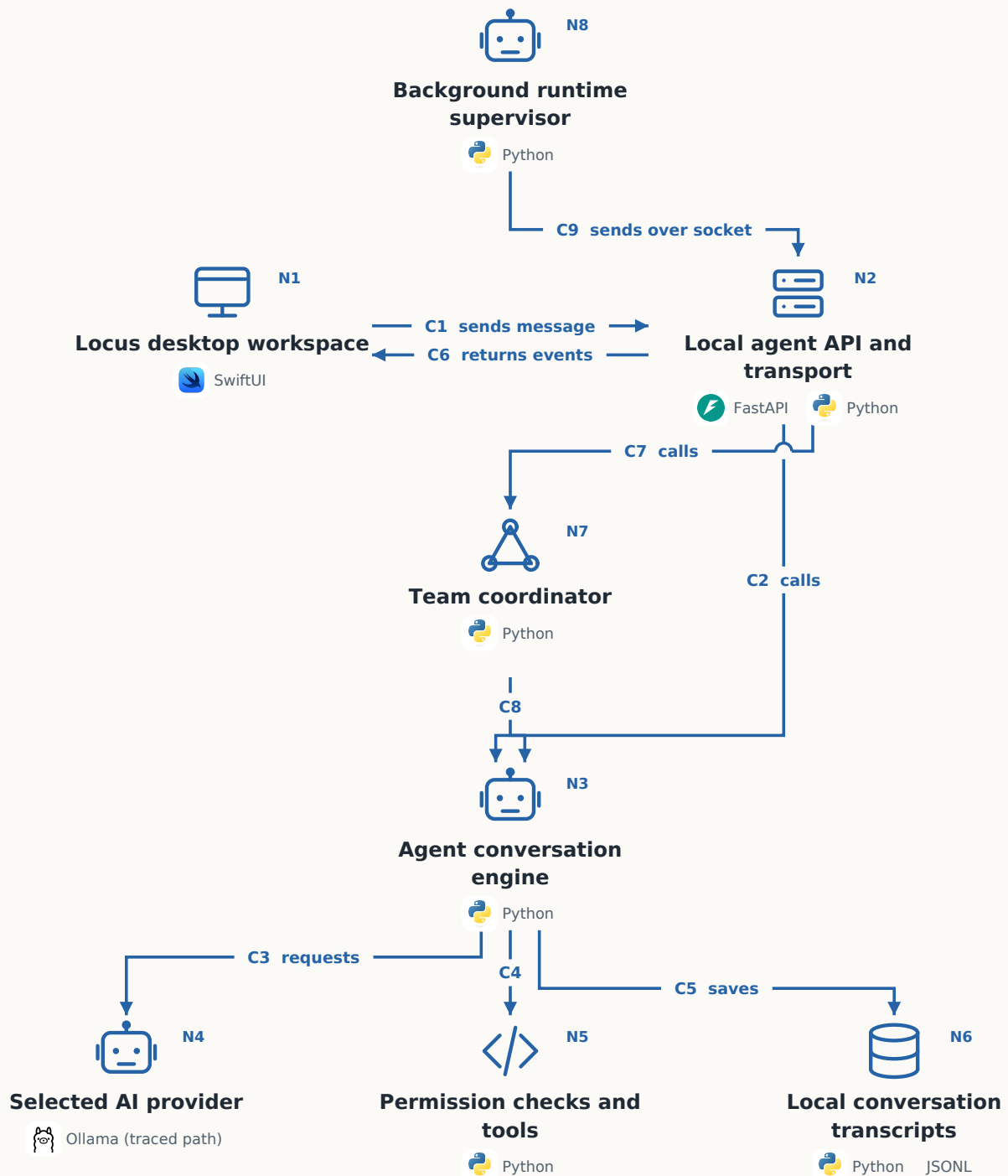
 Recognized source files in the supplied repository inventory, including tests and tools. Physical lines include comments and blank lines; this is code volume, not importance. Excludes: Ignored or absent inventory entries; Vendor, third-party, bundled skills, build and generated directories; named bundles and minified JS/CSS; Manifests, documentation, data, media and other non-source formats; Secret files, symlinks, unreadable/binary files and source files over 8 MB (minified means a JS/CSS line over 20k bytes).

How do the parts connect?

SYSTEM MAP / Arrows show recorded relationships, not execution order

The native workspace talks to the agent backend. Conversation work reaches a selected model, permitted tools and local transcripts; team and recurring work reuse execution paths. This small view omits feature panels, schedule claiming, team specialists/reviewers and many return edges. It shows connections, not one execution order.

Confirmed | E13, E16-E18, E34, E56-E90



Click a part for its files. N = part / C = connection / Blue: confirmed / Amber + dashed: uncertain

Full details: [report.md](#) / [graph.json](#) | E = report evidence

[Code >](#) [System >](#) [Capabilities >](#) [Key parts >](#) [Changes >](#) [Files >](#)

Locus: what can it do?

THE WHOLE PICTURE / A map of capabilities, before the walkthroughs

17 capabilities identified / 3 selected walkthroughs

A local workspace for conversations, project tools, agent teams and recurring work, with optional native integrations, mobile access and remote execution.

Branches group capabilities. They do not show an execution order.



Go deeper: \$x-ray focus "board" | Full inventory and evidence: report.md

Important parts and their files

START HERE / What to understand first, and where it lives

These starting points explain the inspected program. Branches show associated source files; they do not show execution order.

01 Locus desktop workspace



Selected / Confirmed / E38, E59, E81

Connects the Mac workspace to each request and turns streamed backend events into visible replies.

[All files \(+7 more\) >](#)



RELATED FILES

- Locus/LocusApp.swift
- Locus/AppModel+SendPipeline.swift
- Locus/AppModel+BackendEvents.swift

02 Local agent API and transport



Selected / Confirmed / E3, E63, E93

Registers backend feature routes and gates incoming chat connections before handing work to services.

[All files \(+6 more\) >](#)



FastAPI



Python

RELATED FILES

- agent/ollama_code/server.py
- agent/ollama_code/api/_init_.py
- agent/ollama_code/api/chat_transport.py

03 Agent conversation engine



Selected / Confirmed / E14, E67, E75, E76

Runs conversations, selects the managed or classic provider path, and repeats model/tool steps on the classic path.

[All files \(+1 more\) >](#)



Python

RELATED FILES

- agent/ollama_code/core.py
- agent/ollama_code/chat_service.py

04 Permission checks and tools



Selected / Confirmed / E17, E72, E73, E97

Controls which proposed actions may run, requests permission when required, and dispatches tools.

[All component files >](#)



Python

RELATED FILES

- agent/ollama_code/core.py
- agent/ollama_code/tools.py
- agent/ollama_code/chat_service.py

05 Run records and checkpoints



Selected / Confirmed / E96, E98, E99

Keeps durable run events and recovery checkpoints in SQLite; chat transcripts live separately.

[All component files >](#)



SQLite

RELATED FILES

- agent/ollama_code/runstore.py

Work with AI: local Ollama path

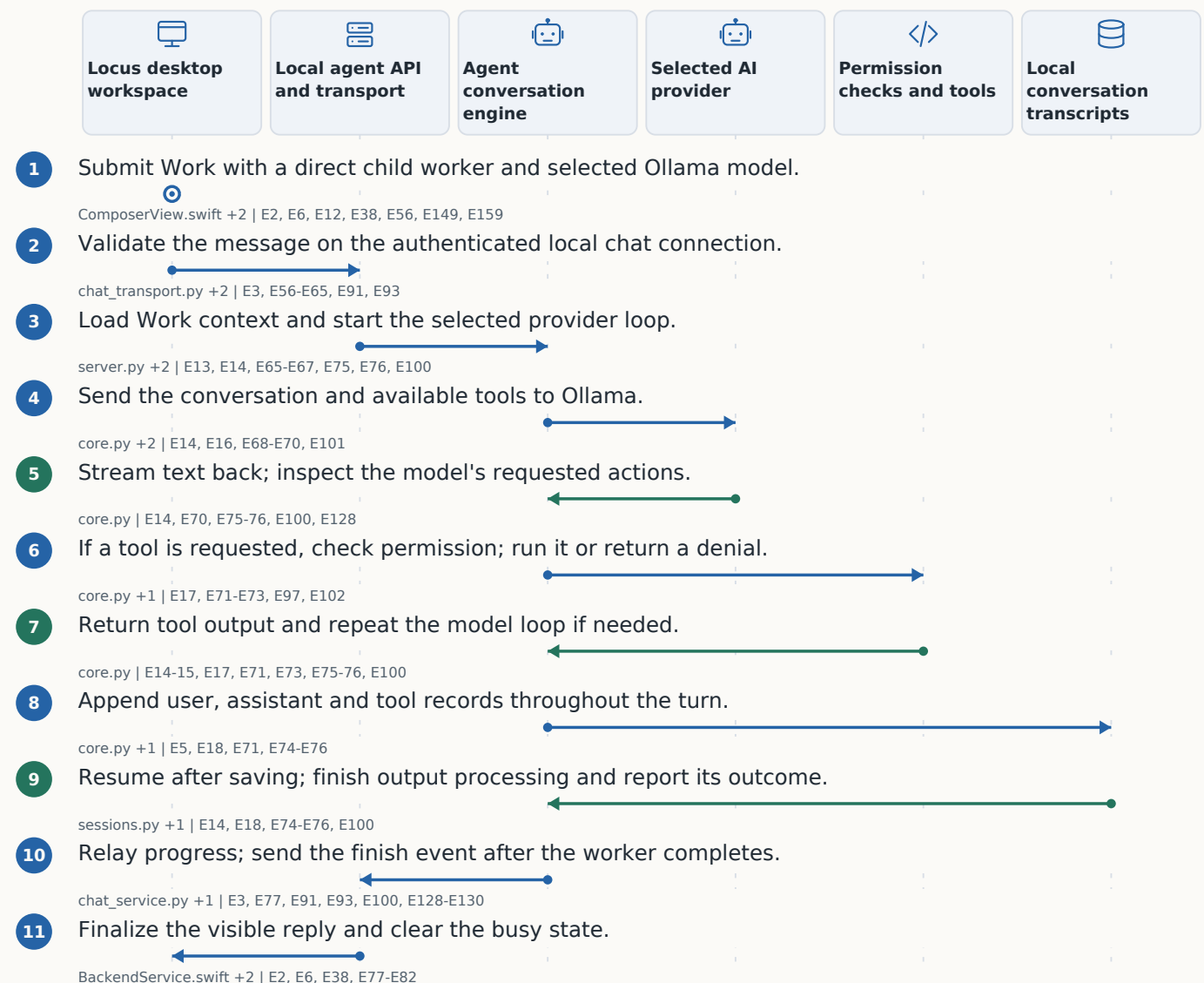
WALKTHROUGH 1 / TRACED WITHIN THE STATED SOURCE SCOPE

Starts when: Send an ordinary Work message with an Ollama model selected and the direct child-worker transport in use.

Example: a direct child worker with local Ollama. Model and permitted tools can repeat; replies stream and records save throughout the turn.

Confirmed | E13, E16, E18, E60, E66, E71, E82

Read downward. Green arrows return; dashed = uncertain. Click a participant for its files.



Give a coding task to a team

WALKTHROUGH 2 / TRACED WITHIN THE STATED SOURCE SCOPE

Starts when: Choose a team in the composer and send a Work message. This representative path uses the Locus-managed engine, ordered coding jobs and a plan containing specialists and a reviewer.

A Locus-managed team follows an approved plan: specialists gather evidence, coding jobs run, and configured reviewers check results. Checkout creation is optional.

Confirmed | E30-E36, E83, E103, E104, E106, E141

Read downward. Green arrows return; dashed = uncertain. Click a participant for its files.



When a recurring agent becomes due

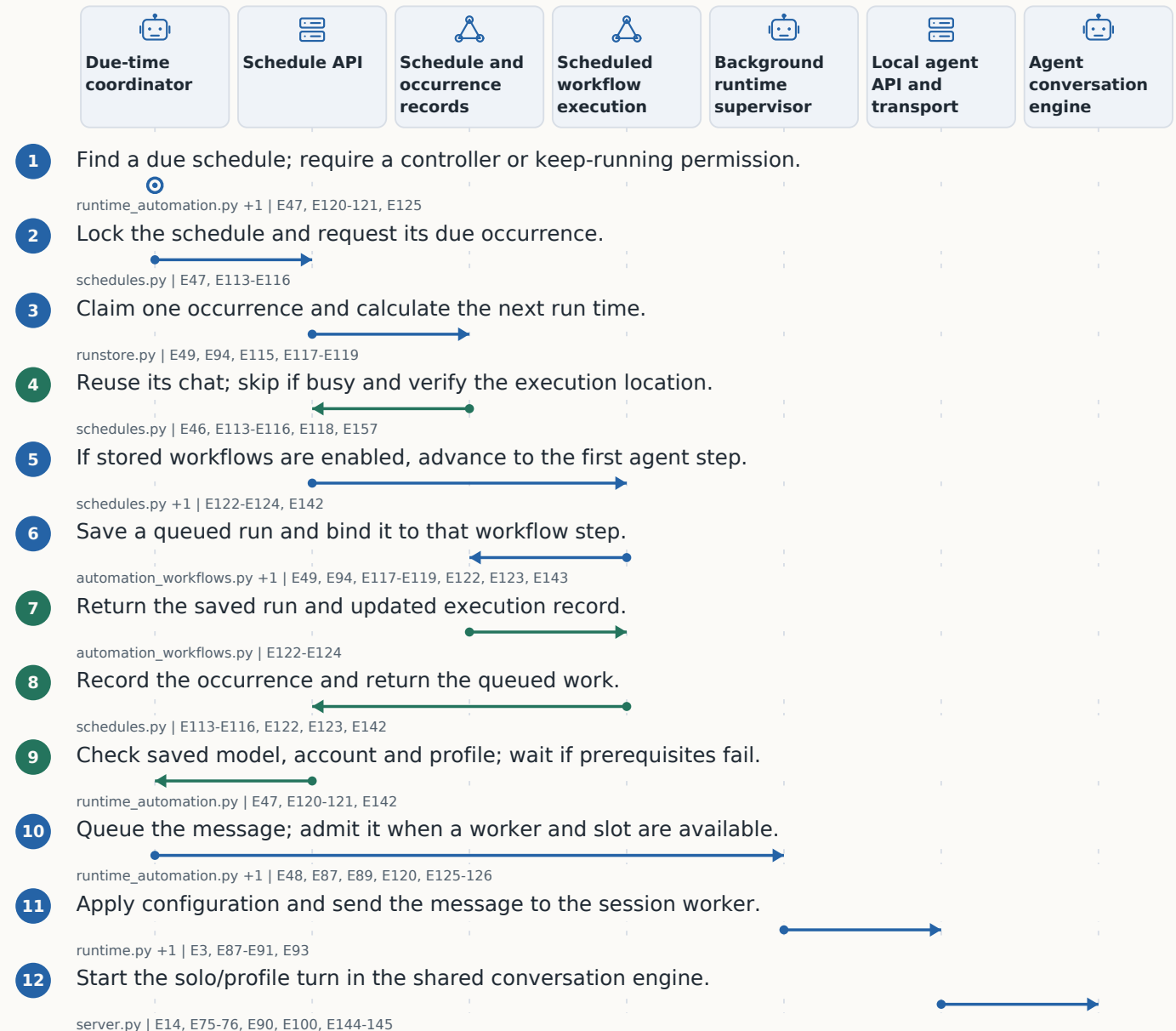
WALKTHROUGH 3 / PARTIAL TRACE - gaps remain

Starts when: An enabled schedule reaches its next run time while the runtime is eligible to run it.

A due schedule claims its slot and queues the first agent step of a stored workflow. This trace ends at agent execution; later completion and delivery remain untraced.

Confirmed | E47, E89, E118, E120, E123, E142, E145

Read downward. Green arrows return; dashed = uncertain. Click a participant for its files.



Where the trace stops

The trace covers the first solo/profile agent step of a stored workflow. Later workflow conditions, approvals, team execution and complete end-to-end outcome reconciliation are not traced here.

Runtime follow-up calls and history queries are evidenced, but actual completion, delivery of notifications and deployed availability were not exercised.

Where would you change it?

CHANGE MAP / Where to start, why, and what to check

File names resolve in the final component map. Possible impact identifies related parts, not guaranteed breakage. Suggested checks have not been run.

IF YOU WANT TO...	START IN THESE FILES	WHY HERE
 Change the workspace panels Possible impact: Locus desktop workspace, Project files and review tools Verify: Open each affected panel and confirm its menu label selects the intended destination.	InspectorView.swift InspectorModels.swift E6, E162	These files register the panels and their visible destinations.
 Change model selection Possible impact: Local agent API and transport, Model and account selection Verify: Switch providers in a disposable profile; confirm native catalog and account state match the server's accepted provider branch.	providers.py ProviderAccountsModel.swift E4, E24	Provider APIs accept changes; native account state owns model catalogs.
 Change extension management Possible impact: Local agent API and transport, Plugins, skills and MCP Verify: Install and remove a disposable extension; verify tool and MCP listings refresh after each mutation.	extensions.py E9, E28	Registered extension operations and their mutations are handled here.
 Change saved-output behavior Possible impact: Locus desktop workspace, Workspace library Verify: Request a revision from a saved version; confirm busy chats or occupied drafts reject it without replacing draft content.	AppModel+LibraryIntegration.swift E8, E37	This joins conversation outputs to saved versions and library actions.
 Change Calendar or Board Possible impact: Locus desktop workspace, Calendar and workspace board Verify: Check that invalid calendar details are rejected and valid entries use a writable calendar; confirm Board changes stay in the selected workspace.	InspectorCalendarTab.swift InspectorBoardTab.swift E19, E23	Separate panel files own Calendar and Board behavior.
 Change what Send includes or when requests queue Possible impact: Local agent API and transport Verify: With the backend offline, verify input stays unsent; while a run is busy, verify a second request queues and preserves its selected mode.	AppModel+SendPipeline.swift E59, E60, E149	This pipeline builds admitted requests; the backend contract must stay in sync.
 Change model/tool iteration or completion behavior Possible impact: Selected AI provider, Permission checks and tools, Local conversation transcripts, Local agent API and transport Verify: On the traced local Ollama branch, verify a tool result returns to the next model step and turn_done ends the turn.	core.py E14, E71, E76, E100	The core repeats model/tool exchanges and decides when a turn ends.
 Change action permissions or built-in tools Possible impact: Agent conversation engine, Locus desktop workspace Verify: For an action requiring approval, compare allow and deny; denial should return a tool result without executing the action.	core.py tools.py chat_service.py E17, E72, E73, E97	These files coordinate permission decisions, tool execution and result text.

Where would you change it?

CONTINUED

CHANGE MAP / Where to start, why, and what to check

IF YOU WANT TO...	START IN THESE FILES	WHY HERE
 Change local chat retention or persistence guarantees	sessions.py core.py E18, E66, E74	Session append rules control ordinary versus strict persistence.
Possible impact: Agent conversation engine, Local agent API and transport Verify: In a disposable profile, make transcript writes fail; verify strict input admission stops before execution while ordinary append follows its recorded fallback.		
 Change how streaming answers appear	AppModel+BackendEvents.swift BackendService.swift E78, E81, E82	Backend events build streamed reply blocks and clear busy state at completion.
Possible impact: Local agent API and transport Verify: Stream a reply and finish it; verify final content replaces partial text and completion clears busy state.		
 Change how a team divides and reviews work	orchestration.py server.py AppModel+TeamOps.swift E32-E34, E106, E139, E152, E161	The orchestrator plans work; the server runs writing/review; the desktop sends decisions.
Possible impact: Locus desktop workspace, Agent conversation engine, Managed task checkout, Run history and checkpoints Verify: Use the Locus-managed ordered path with a reviewer; verify dependency order and that missing verdicts or exhausted repairs pause completion.		
 Change where delegated coding work runs	worktrees.py server.py E107, E108, E160	Worktree setup creates the private checkout; the team runner enters it when enabled.
Possible impact: Agent conversation engine, Run history and checkpoints Verify: Enable managed isolation in a Git workspace with no active task; verify execution enters the new private checkout. Trace source-workspace apply separately.		
 Change which recurrence patterns an agent can use	ollama_code/schedules.py AppModel+RunQueueAndActivity.swift E45, E118, E156, E163	Recurrence logic and desktop checks must agree on the next-run time.
Possible impact: Due-time coordinator, Schedule API Verify: Compare valid and rejected recurrence drafts; verify native checks and backend agree on a future next-run time.		
 Change how overlapping scheduled runs are handled	api/schedules.py runstore.py E115, E118, E157	Schedule dispatch skips busy chats; RunStore makes occurrence claims durable.
Possible impact: Due-time coordinator, Background runtime supervisor Verify: Dispatch while the scheduled chat is busy; verify a recorded skip and one-time rearming without duplicate occurrence claims.		
 Change when scheduled work can keep running	runtime_automation.py runtime.py E47, E125, E154	Both due dispatch and admission check controller/keep-running eligibility.
Possible impact: Schedule API, Local agent API and transport Verify: Disconnect with keep-running off and on; verify admission follows that policy and retains one writer per workspace. Final workflow reconciliation remains untraced.		

COVERAGE MAP / Follow the source paths to their evidence boundary

17 mapped capabilities → 3 selected walkthroughs below

Menu and panel entryptoints are mapped across the native workspace and backend. Three distinct journeys were selected for deeper tracing.

! OUTSIDE THIS TRACE / Every panel action and provider variation. + 3 other limits in report.md

The local Ollama/direct-worker branch connects submission, streaming, permitted tools, JSONL records and the visible reply.

START / Locus desktop workspace

LAST TRACED / Locus desktop workspace

SCOPED END

! OUTSIDE THIS TRACE / Independent-runtime relay internals and restart/reconnect replay. + 4 other limits in report.md

Confirmed source evidence | E13, E16, E18, E60, E66, E71, E72, E82

The scoped team path covers plan approval, specialists, ordered coding, configured review and a handoff. Run metadata uses SQLite, separately from chat transcripts.

Diagram illustrating a sequence of nodes (1 to 12) connected by arrows, representing a workflow or process flow. The nodes are labeled as follows:

- Node 1: START / Locus desktop workspace
- Node 2: (unlabeled)
- Node 3: (unlabeled)
- Node 4: (unlabeled)
- Node 5: (unlabeled)
- Node 6: (unlabeled)
- Node 7: (unlabeled)
- Node 8: (unlabeled)
- Node 9: (unlabeled)
- Node 10: (unlabeled)
- Node 11: (unlabeled)
- Node 12: LAST TRACED / Locus desktop workspace

A vertical line after node 12 is labeled **SCOPED END**.

! OUTSIDE THIS TRACE / Live model/provider responses and end-to-end task success. + 3 other limits in report.md

Source checks cover creation, recurrence, durable claims, first-step admission and history. Later workflow completion is only partly traced.

Diagram illustrating a trace sequence:

```

graph LR
    1((1)) --> 2((2))
    2 --> 3((3))
    3 --> 4((4))
    4 --> 5((5))
    5 --> 6((6))
    6 --> 7((7))
    7 --> 8((8))
    8 --> 9((9))
    9 --> 10((10))
    10 --> 11((11))
    11 --> 12((12))
    12 -.-> PT[?]
  
```

The sequence consists of nodes 1 through 12, connected by arrows. Nodes 1 through 10 are labeled "START / Due-time coordinator" and "LAST TRACED / Agent conversation engine" respectively. Nodes 11 and 12 are followed by a dashed line and a question mark, labeled "PARTIAL TRACE".

TRACE BOUNDARY / The trace covers the first solo/profile agent step of a stored workflow. Later workflow conditions, approvals, team execution and complete end-to-end outcome reconciliation are not traced here. All additional gaps and limits are in report.md.

Confirmed source evidence | E47, E49, E89, E111, E118, E120, E123, E142, E145

What would we check next?

NEXT-CHECK MAP / Turn each open question into a focused investigation

Follow one question to its next check. These are gaps in this report's evidence, not identified product defects.

 Configured environment Which accounts, runtimes, integrations and native permissions are enabled on a real installation?	→	 Inspect the relevant settings and one focused capability in an explicitly scoped live check.
 Other chat execution branches Which provider and worker transport are active for a particular user request?	→	 For a focused follow-up, inspect the chosen route and its relevant adapter/relay using non-secret configuration and then trace that branch.
 Live team execution Does the selected team complete a real task with its configured providers and permissions?	→	 Use a disposable project and a representative team to verify plan review, a permitted coding change, reviewer output and final task changes.
 Alternative delegation engines and writer layouts Do hosted delegation and parallel-writer integration preserve the same boundaries on all error/recovery paths?	→	 Focus separately on the selected engine, parallel writer worktree integration, or resume/retry behavior before claiming full coverage.
 Writer permission enforcement How are every writer tool and provider-native execution route restricted by the active access ceiling?	→	 Trace AgentCore, tool policy and native-provider execution for each active permission mode.
 Runtime availability Is the user's actual runtime installed, running and provisioned to execute this schedule when the desktop is closed?	→	 If operational confirmation is needed, inspect the app's runtime and scheduled-agent status with the user's intended schedule.
 Scheduled workflow completion How does each configured later step finish, and what exact result appears for the user?	→	 Focus on the chosen agent's workflow and trace completion events through run state, workflow advancement and inspector presentation.

Components and their files

FILE MAP / Every mapped part, with its associated source files

Files sit beneath their component. 'From' lists incoming connections by directory number; look there for the connected part's files. Component links do not imply direct file imports. Full file evidence: graph.json.

Folder key: [1]/ = Locus/ | [2]/ = agent/ollama_code/

01 Locus desktop workspace



- [1]/LocusApp.swift
- [1]/InspectorView.swift
- [1]/BackendService.swift
- [1]/ComposerView.swift
- [1]/AppModel+SendPipeline.swift
- [1]/AppModel+ChatWorkers.swift
- [1]/TaskWorkerRuntime.swift
- [1]/AppModel+BackendEvents.swift
- [1]/AppModel+TeamOps.swift
- [1]/TeamRunLiveModel.swift

From: 02, 10, 17 | E2, E6, E38

02 Local agent API and transport



- [2]/server.py
- [2]/api/__init__.py
- [1]/BackendProcess.swift
- [1]/AppModel+ChatWorkers.swift
- [1]/BackendService.swift
- [2]/api/chat_transport.py
- [2]/chat_transport_runtime.py
- [2]/chat_service.py
- [1]/AppModel+SendPipeline.swift

From: 01, 13, 27 | E3, E91, E93

03 Project files and review tools



- [1]/AppModel+WorkspaceBrowser.swift
- [1]/InspectorView.swift
- [1]/Models/InspectorModels.swift

From: 01 | E6, E38, E43

04 Shared browser



- [1]/InspectorBrowserTab.swift

From: 01 | E20, E21

05 Model and account selection



- [1]/ProviderAccountsModel.swift
- [2]/api/providers.py

From: 02 | E4, E24, E25

06 Plugins, skills and MCP



- [2]/api/extensions.py

From: 02 | E9, E28

07 Goals and saved task plans



- [1]/AppModel+Goals.swift
- [2]/api/goals.py
- [2]/api/capsules.py

From: 01 | E7, E39, E40, E53

08 Workspace library



- [1]/AppModel+LibraryIntegration.swift

From: 01 | E8, E37

09 Calendar and workspace board



- [1]/InspectorCalendarTab.swift
- [1]/InspectorBoardTab.swift

From: 01 | E6, E19, E22, E23

10 Mobile companion access



- [1]/CompanionSettingsView.swift
- [1]/AppModel+MobileCompanion.swift

From: none recorded | E10, E41, E42

11 Remote runtime controls



- [1]/RemoteRuntimesView.swift
- [2]/api/runtime_deploy.py

From: none recorded | E11, E50, E51

12 Computer, voice and simulator controls



- [1]/ComputerControlService.swift
- [1]/VoiceComposerControls.swift
- [1]/InspectorSimulatorTab.swift

From: 01 | E26, E27, E54, E55

13 Agent conversation engine



- [2]/core.py
- [2]/chat_service.py
- [2]/server.py

From: 02, 14, 15, 16, 17 | E14, E75, E76, E100

14 Selected AI provider



- [2]/core.py
- [2]/model_usage.py
- [2]/ollama.py

From: 13 | E14, E16, E68, E70, E101

15 Permission checks and tools



- [2]/core.py
- [2]/tools.py
- [2]/chat_service.py

From: 13 | E17, E72, E73, E97, E102

16 Local conversation transcripts



- [2]/sessions.py
- [2]/core.py

From: 13 | E5, E18, E74

17 Team coordinator



- [2]/orchestration.py
- [2]/server.py
- [1]/AppModel+TeamOps.swift
- [2]/chat_service.py

From: 01, 02, 13, 18, 19, 20 | E32, E36, E85, E103

18 Read-only specialists



- [2]/orchestration.py

From: 17 | E33, E104, E105

19 Configured reviewers



- [2]/orchestration.py
- [2]/server.py

From: 17 | E35, E106

20 Managed task checkout



- [2]/worktrees.py

From: 17 | E107, E108

21 Run records and checkpoints



- [2]/runstore.py

From: 17, 24 | E84, E95, E96, E98, E99, E109

22 Scheduled agent controls



- [1]/WorkspaceView.swift
- [1]/ScheduleModel.swift
- [1]/AgentInspectorModel.swift
- [1]/AppModel+RunQueueAndActivity.swift

From: 23 | E44, E110-E112

23 Schedule API



- [2]/api/schedules.py

From: 22, 24, 25, 26 | E113-E116

24 Schedule and occurrence records



- [2]/runstore.py
- [2]/agent_inspector_store.py
- [2]/schedules.py

From: 23, 26 | E49, E94, E117-E119

25 Due-time coordinator



- [2]/runtime_automation.py

From: 23, 27 | E47, E120, E121

26 Scheduled workflow execution



- [2]/api/automation_workflows.py

From: 23, 24 | E122-E124

27 Background runtime supervisor



- [2]/runtime.py

From: 25 | E48, E87, E89, E125, E126